News Topic Classification Using TF-IDF, Skip-gram, Machine Learning, and Neural Network Models

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***Abstract*—This project investigates supervised news topic classification using a dataset already divided into training and testing CSV files. The training data were used for exploratory data analysis, preprocessing, model training, validation, and hyperparameter tuning, while the testing data were reserved strictly for final evaluation. Three text-processing settings were evaluated: no preprocessing, extreme preprocessing, and optimum preprocessing. TF-IDF features were used to establish a strong machine learning baseline via Multinomial Naive Bayes and a Deep Neural Network, while Skip-gram Word2Vec embeddings were leveraged alongside recurrent architectures (SimpleRNN, GRU, LSTM, and their bidirectional variants). Performance was compared using accuracy and macro-F1 scores. The final results demonstrate how preprocessing intensity and representation choice affect classification performance, revealing that Skip-gram embeddings with gated architectures yield the highest overall scores, even with minimal preprocessing.**

***Index Terms*—Text classification, TF-IDF, Skip-gram, Word2Vec, RNN, GRU, LSTM, preprocessing, news classification.**

I. Introduction

Text classification is a fundamental natural language processing task in which documents are automatically assigned to predefined categories. In this project, news headlines are classified into four topic classes: Business, Science and Technology, Sports, and World News. The objective is to compare classical machine learning and neural network approaches under different preprocessing strategies and word representation techniques.

The motivation behind this experiment is to understand how preprocessing, feature representation, and model architecture influence classification performance on short news texts. Since headlines are concise and topic-focused, both sparse statistical representations such as TF-IDF and dense distributed representations such as Skip-gram are suitable candidates for experimentation.

II. Methodology

A. Dataset

The dataset contains two CSV files: a training file and a testing file. The text column is *News Headline*, and the target column is *News Topic*. The original training set contains 88,829 rows, while the test set contains 12,000 rows. During EDA, 28,833 exact duplicate rows were found in the training set. These duplicates were removed before creating the validation split to reduce the risk of validation score inflation caused by identical samples appearing in both training and validation sets. The final deduplicated training set contains 59,996 rows.

TABLE I
Dataset Summary

Split	Rows	Duplicate Rows	Classes
Original Training	88,829	28,833	4
Deduplicated Training	59,996	0	4
Testing	12,000	0	4

The test set is perfectly balanced, with 3,000 samples per class. The deduplicated training set remains moderately imbalanced, with Sports and World News appearing more frequently than Business and Science and Technology. Therefore, macro-F1 was used as a major evaluation metric because it gives equal importance to all classes.

TABLE II
Class Distribution After Training Duplicate Removal

Class	Count	Percentage
Sports	19,831	33.05%
World News	18,617	31.03%
Business	11,801	19.67%
Science and Technology	9,747	16.25%

EDA also showed that the text contains HTML tags and repeated wrapper text such as *News Headlines:*. After removing HTML tags for analysis, the average headline length was approximately 38 words in both training and test sets.

B. Preprocessing

Three versions of the dataset were prepared:

1) No Preprocessing: In this setting, the raw text was used directly before vectorization. This version was included to measure how well the models perform without cleaning HTML artifacts or normalizing the text.

2) Extreme Preprocessing: Extreme preprocessing included HTML removal, lowercasing, URL removal, punctuation removal, number removal, stopword removal, lemmatization, and stemming. This setting was used to test whether aggressive normalization improves or reduces classification performance.

3) Optimum Preprocessing: Optimum preprocessing was selected based on EDA. It removed HTML tags, repeated wrapper text, URLs, and unnecessary symbols, while preserving numbers, apostrophes, and useful topic words. Stemming and stopword removal were avoided in the optimum version because named entities and important topical words may be useful for classifying news headlines.

C. Word Representation Techniques

Two representation techniques were used:

1) TF-IDF: TF-IDF converts documents into sparse vectors based on term importance. It was used with Multinomial Naive Bayes and the deep neural network. Hyperparameters such as maximum features, n-gram range, and smoothing alpha were tuned using validation macro-F1.

2) Skip-gram Word2Vec: Skip-gram Word2Vec was trained on the training split only. The learned embeddings were used as the embedding layer for all recurrent neural network models. Hyperparameters such as hidden units, dropout, and learning rate were tuned using validation macro-F1.

III. Results and Discussion

Table III, along with the visual comparisons in Figure 1 and Figure 2, details the comprehensive performance of all 24 evaluated combinations across preprocessing techniques, word representations, and model architectures.

Analyzing the preprocessing strategies reveals an unexpected but insightful trend: *no preprocessing* consistently yielded highly competitive, and often superior, results. The overall best model—the GRU trained on raw text—achieved a test accuracy of 91.83% and a macro-F1 score of 0.9180. This suggests that the Skip-gram embeddings and recurrent architectures successfully learned to handle or bypass HTML artifacts autonomously, mitigating the need for aggressive cleaning that might inadvertently strip away valuable surface-level context. To visualize this effectiveness, Figure 3 presents the confusion matrix for this top-performing model, demonstrating strong true-positive rates across all four distinct classes.

When comparing word representation techniques, the dense semantic embeddings of Skip-gram visibly outperformed the sparse TF-IDF approach in almost all deep learning scenarios. While the best TF-IDF model (DNN with extreme preprocessing) reached a strong macro-F1 of 0.9116, the Skip-gram representations paired with gated architectures continually pushed scores into the 0.914–0.918 range.

Finally, while bidirectional models (BiGRU and BiLSTM) performed excellently (e.g., Optimum BiGRU at 0.9178 Macro-F1), they did not provide a drastic leap over their unidirectional counterparts. The standard GRU's peak performance indicates that left-to-right context alone, when adequately parameterized, was highly sufficient for these concise headlines. Furthermore, the significant collapse of the SimpleRNN model's performance under extreme and optimum preprocessing highlights its sensitivity to sequence disruption. This severe degradation is clearly illustrated in the confusion matrix in Figure 4, which reveals that the worst-performing model completely failed to discriminate between classes, defaulting its predictions entirely to a single class (Sports). This emphasizes the well-documented vanishing gradient limitations of vanilla RNNs compared to GRU and LSTM cells.

IV. Conclusion

This project successfully evaluated multiple preprocessing strategies, word representations, and classification models for multi-class news topic classification. The experiments definitively demonstrate that Skip-gram embeddings combined with gated recurrent architectures (specifically the GRU) yield the highest classification accuracy. Additionally, the results strongly suggest that for sequence models utilizing learned embeddings, preserving the raw text—rather than applying extreme normalization or filtering—can actually enhance model performance by retaining implicit contextual cues.

Limitations of this study include the computational overhead of training extensive bidirectional neural models on longer sequences, given that headlines are inherently short and may not always necessitate heavy sequence modeling. Future work may explore the integration of Transformer-based models such as BERT, the use of large-scale pre-trained embeddings (e.g., GloVe or FastText), and deeper error analysis to fine-tune extraction heuristics for consistently misclassified topics.

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TABLE III
FINAL MODEL COMPARISON

Preprocessing	Representation	Model	Best Hyperparameters	Test Acc.	Test F1
No preprocessing	TF-IDF	MultinomialNB	α=0.2, max_feat=30k	0.9007	0.9001
Extreme preprocessing	TF-IDF	MultinomialNB	α=0.2, max_feat=30k	0.9044	0.9039
Optimum preprocessing	TF-IDF	MultinomialNB	α=0.2, max_feat=30k	0.9007	0.9001
No preprocessing	TF-IDF	DNN	max_feat=15k, ngram=(1,2)	0.9050	0.9050
Extreme preprocessing	TF-IDF	DNN	max_feat=15k, ngram=(1,2)	0.9118	0.9116
Optimum preprocessing	TF-IDF	DNN	max_feat=12k, ngram=(1,1)	0.9101	0.9097
No preprocessing	Skip-gram	SimpleRNN	units=256, drop=0.3	0.8908	0.8898
Extreme preprocessing	Skip-gram	SimpleRNN	units=512, drop=0.5	0.2683	0.1470
Optimum preprocessing	Skip-gram	SimpleRNN	units=256, drop=0.3	0.2500	0.1000
No preprocessing	Skip-gram	GRU	units=512, drop=0.5	**0.9183**	**0.9180**
Extreme preprocessing	Skip-gram	GRU	units=256, drop=0.3	0.9096	0.9091
Optimum preprocessing	Skip-gram	GRU	units=256, drop=0.3	0.9144	0.9143
No preprocessing	Skip-gram	LSTM	units=256, drop=0.3	0.9143	0.9141
Extreme preprocessing	Skip-gram	LSTM	units=256, drop=0.3	0.9106	0.9102
Optimum preprocessing	Skip-gram	LSTM	units=256, drop=0.3	0.9105	0.9100
No preprocessing	Skip-gram	BiRNN	units=256, drop=0.3	0.8960	0.8956
Extreme preprocessing	Skip-gram	BiRNN	units=256, drop=0.3	0.9006	0.8996
Optimum preprocessing	Skip-gram	BiRNN	units=256, drop=0.3	0.8970	0.8965
No preprocessing	Skip-gram	BiGRU	units=256, drop=0.3	0.9175	0.9172
Extreme preprocessing	Skip-gram	BiGRU	units=256, drop=0.3	0.9148	0.9144
Optimum preprocessing	Skip-gram	BiGRU	units=256, drop=0.3	0.9180	0.9178
No preprocessing	Skip-gram	BiLSTM	units=256, drop=0.3	0.9124	0.9119
Extreme preprocessing	Skip-gram	BiLSTM	units=256, drop=0.3	0.9168	0.9164
Optimum preprocessing	Skip-gram	BiLSTM	units=256, drop=0.3	0.9157	0.9151

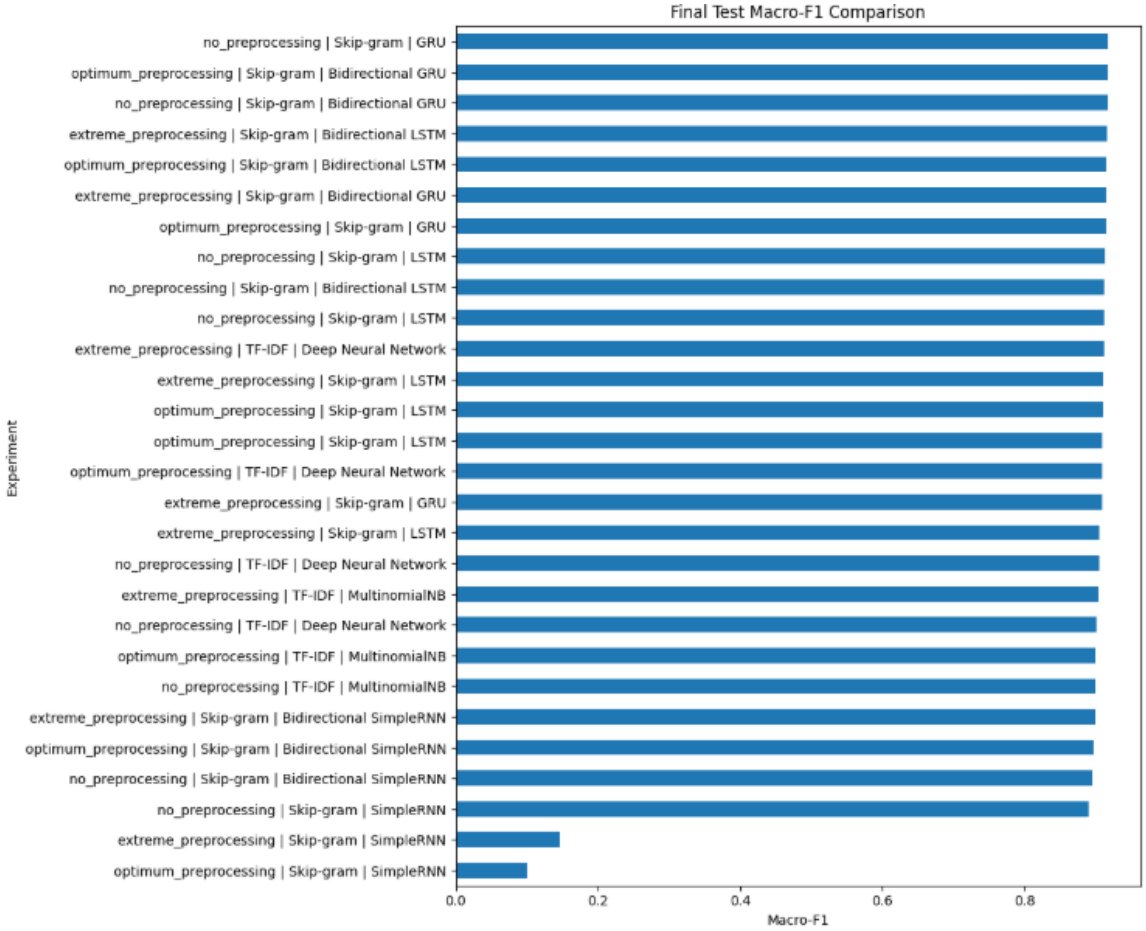


Fig. 1. Final Test Macro-F1 Comparison across all models and preprocessing techniques.

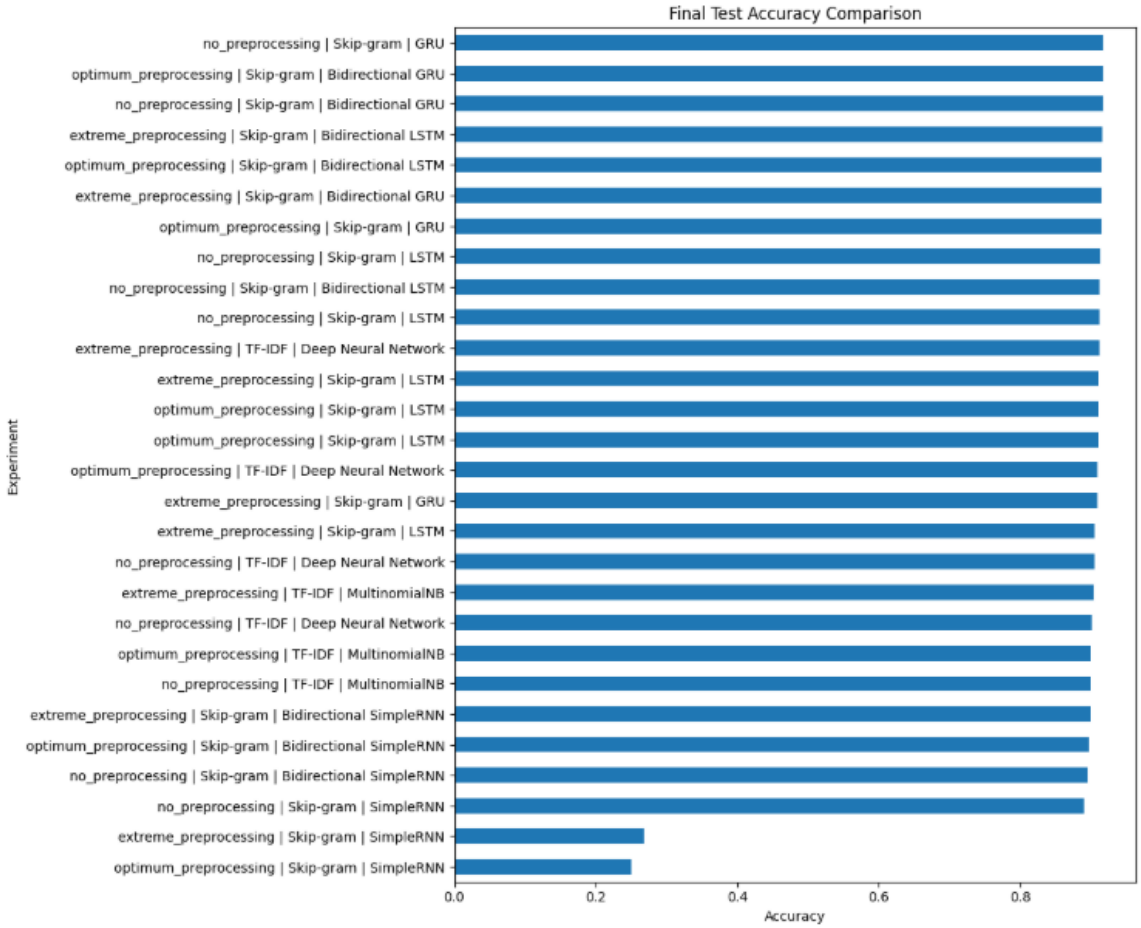


Fig. 2. Final Test Accuracy Comparison across all models and preprocessing techniques.

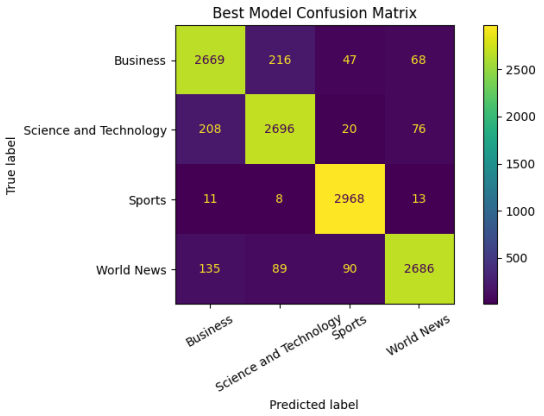


Fig. 3. Confusion Matrix for the Best Performing Model (GRU with No Preprocessing).

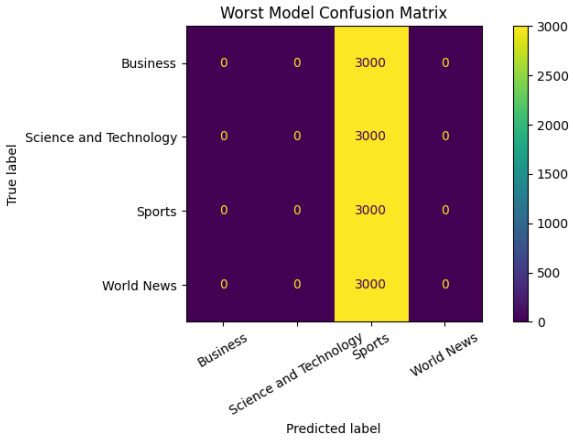


Fig. 4. Confusion Matrix for the Worst Performing Model (SimpleRNN with Optimum Preprocessing).